

Good jobs and bad jobs for Indonesia's informal workers
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Abstract

There is consistent evidence that informal workers face a plethora of poor outcomes, ranging from a lack of social protection, poor working conditions, and low average incomes. However, these findings are based on the assumption that informal workers are homogeneous. In this paper, we deviate from this view by accounting for the possibility of “winners” and “losers” among informal workers. Using unconditional quantile regressions with fixed effects, we are able to estimate the formal-informal wage gap at different segments of the earnings distribution. While the evidence does confirm the existence of a substantial earnings penalty for many informal workers, there is also a group that earns roughly similar incomes as formal workers. The evidence is consistent with Fields' (1990; 2005) model of a heterogeneous informal sector.

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1 Introduction

There is no shortage of evidence on the disadvantages faced by informal workers in the labour market. Decades of research on this subject have yielded a plethora of poor outcomes, ranging from a lack of social protection, poor working conditions, and low average incomes (ILO, 2002; 2013). It is therefore not surprising that informal employment has often been used as a proxy for bad jobs (Pages & Madrigal, 2008). But is this view truly justified? For one, the frequently documented problems associated with informality may be offset by non-pecuniary benefits such as flexibility and autonomy (Fields, 2005; Perry, et al., 2007; World Bank, 2010). Alternatively, informal work may allow individuals to capitalize on their comparative advantages in the labour market, thereby resulting in the highest possible income given their skills and characteristics (Perry, et al., 2007; Arias & Khamis, 2008).

More importantly, not all informal jobs can be considered “bad jobs” insofar as earnings are concerned. As Kucera and Roncolato (2008, p. 322) argued, “some informal jobs are better than some formal jobs with respect to incomes.” This study aims to shed light on the heterogeneity of informal employment in Indonesia. This country provides a good case study for a number of reasons. Similar to other developing countries, a substantial portion of its workforce is employed in informal jobs. The ILO (2018) estimates that informal workers in Indonesia account for more than 80 percent of total employment. Second, its labour regulations – which are among the most rigid in the world – are believed to have worsened the segmentation between formal and informal workers (Matsumoto & Verick, 2011). Finally, Indonesia remains one of the few developing countries with large-scale longitudinal data.

Contrary to the majority of empirical studies, we explicitly account for the possibility that informal workers are comprised of both “winners” and “losers”. We achieve this by estimating wage gaps at various unconditional quantiles as proposed by Firpo, Fortin and Lemieux (2007, 2009) and extended by Borgen (2016). This approach yields considerable differences from the standard findings in the literature, which have primarily focused on average earnings. While the evidence does confirm the existence of a sizable earnings penalty for some informal workers, there is also a group that earns roughly similar incomes as formal workers. These findings, which are in line with Fields’ (1990) theory, highlight the fact that informal employment may indeed be an optimal choice for some individuals.

The paper proceeds as follows: Section 2 provides the theoretical framework followed by a brief review of the empirical literature in Section 3. Section 4 briefly outlines the data and discusses the main methodologies used in the study. The results are then presented and discussed in Section 5. Section 6 supplements the main analysis by providing a number of robustness checks, and Section 7 concludes.

2 Theoretical framework

Informal employment, broadly defined as work that is not regulated and protected by labour legislation (*see section 4*), has traditionally been viewed as the result of a dualistic or segmented labour market (Lewis, 1954; Harris & Todaro, 1970; Tokman, 1992). Segmentation may arise either due to barriers to entry (e.g. geographic, information) in the formal sector, or because of sectoral differences in wage-setting and employment policies (Dickens & Lang, 1992). The latter includes mechanisms that push wages above the market equilibrium. One of the most commonly cited examples are labour market institutions such as minimum wages, employee dismissal regulations, and trade unions (Fields, 2005). Alternatively, firms may offer *efficiency wages* – wages above the equilibrium level – to increase profits by inducing greater productivity among workers, increasing retention, and reducing costs associated with

turnover (e.g. Leibenstein, 1957; Stiglitz, 1976; Akerlof & Yellen, 1986). Both instances raise the price of labour, thereby leading to a reduction in the demand for workers in the formal sector.

More broadly, dualism may reflect a country's level of development and economic structure. Informal employment is naturally higher in economies where the "traditional" sector (i.e. agriculture) dominates. But it may also persist even as countries develop, particularly when the economy shifts into low-productivity services which are more predisposed towards informality than manufacturing (Perry, et al., 2007). Segmentation may also be reinforced by external factors such as globalization. For example, changes in production structures have allowed firms to outsource their production to suppliers abroad, some of whom are in informal enterprises. Likewise, the increased global competition can induce firms to adopt cost minimization strategies which include hiring workers on non-standard arrangements or laying off formal workers who are then forced to take on informal jobs (ILO, 2013a; Goldberg & Pavcnik, 2003).

The dualistic or segmented view directly contradicts theories of competitive labour markets, which depict informal employment as a voluntary choice (e.g. Magnac, 1991; Pradhan & van Soest, 1995; Maloney, 1999). These models imply that at least for some individuals, being an informal worker is an optimal or equilibrium outcome. This may arise when workers trade pecuniary rewards for non-pecuniary characteristics (i.e. compensating differentials) or when they sort into jobs based on their comparative advantage (Roy, 1951). The latter may manifest through sorting or selection based on some observed or unobserved characteristic. For example, workers with entrepreneurial skills may choose self-employment over wage or salary work. Similarly, some individuals may opt out of the formal sector if their skills are better rewarded in jobs that tend to be informal (e.g. construction) (Perry, et al., 2007).

These seemingly contradictory views need not be mutually exclusive. Conceptually, Fields (1990, 2005) reconciled the two in his model of a heterogeneous informal sector.² In his words, the informal sector itself is divided into two distinct segments – a free or "easy-entry" tier and an "upper-tier". The former is composed of individuals who rely on informal jobs as a means of avoiding unemployment. As the name suggests, this segment has minimal barriers to entry and is open to virtually anyone seeking work. This is in sharp contrast to the upper tier, where workers are generally more skilled or have accumulated financial capital to start their own small enterprises. In this case, the lower tier is consistent with segmentation while the upper tier is more in line with the competitive labour market model.

The three main models of the labour market (i.e. segmented, competitive, and heterogeneous) result in different predictions regarding the earnings of informal workers. If the labour market is segmented, informal workers should earn less compared to formal workers who share identical characteristics. Earnings are therefore determined by the nature of employment (i.e. formal or informal), and not by characteristics or abilities per se. However, a significant earnings penalty for informal workers is not a sufficient condition for segmentation. That is, informal workers may face an earnings disadvantage even when the labour market is perfectly competitive. This may result from differences in human capital (i.e. comparative advantage) or from non-pecuniary factors which earnings alone cannot capture (i.e. compensating differentials).

² The 15th International Conference of Labour Statisticians defined the informal sector as a "group of production units which [...] form part of the household sector as household enterprises, or equivalently, unincorporated enterprises owned by households." However, this narrow definition was found to be an "inadequate, if not misleading, term to reflect the dynamic, heterogeneous, and complex aspects of a phenomenon which is not, in fact, a 'sector' in the sense of a specific industry group or economic activity." (ILO, 2002, p. 2). A new term was thus created – informal employment – which defines informality from the perspective of jobs. The present study, with its focus on informal workers, fits the latter perspective.

Although comparative advantage is most commonly associated with a wage premium, it may also manifest as a penalty due to differences in workers' skills or abilities. It is therefore necessary to account for the effect of individual characteristics in order to distinguish between segmentation and comparative advantage. Alternatively, wage penalties may reflect compensating differentials if informal workers accept lower pay in exchange for the non-pecuniary characteristics (e.g. flexibility, autonomy) of the job. If this is the case, informal workers should report the same overall utility as formal workers – an issue that is indirectly addressed in the robustness checks. Against this background, some scholars have argued that focusing on wage gaps alone does not provide conclusive evidence regarding segmentation (e.g. Rosenzweig, 1988; Maloney, 1999). Nevertheless, examining earnings yields important information particularly in cases where informal workers earn similar or even greater incomes than formal workers.

Given differences in preferences and abilities, labour markets are likely to exhibit some combination of the above scenarios. This would be consistent with Fields' (1990, 2005) heterogeneous model, which predicts that the size of the earnings gap varies along the income distribution. While one segment may face a substantial earnings penalty, the gap may be small or negligible for the other segment. Alternatively, there may be a penalty in one segment and a premium in the other segment. In the next section, we review previous evidence on the topic and highlight the gaps that this study aims to address.

3 Review of literature

The empirical literature has relied heavily on income comparisons to test for segmentation between formal and informal workers. The findings have "almost universally" (Gindling, Mossaad, and Newhouse, 2016, p. 6) been in favor of the former with some notable exceptions (e.g. Pratap & Quintin, 2006; Staneva & Arabsheibani, 2014). These studies, which are largely based on the experience of lower- and upper-middle income countries, share the underlying assumption that competitive labour markets imply "equal pay for equal skills" (Perry, et al., 2007, p. 87). That is, differences in abilities should fully explain the earnings gap rather than the sector of employment. Testing this hypothesis has been a major focus of empirical studies, with various estimation strategies being applied on both cross-sectional and longitudinal datasets. In this section, we briefly review the methodologies and corresponding findings that have shaped the literature thus far.

One strand of literature has focused on deriving unbiased estimates of the sectoral coefficient. The objective of these studies is to control for both observed and unobserved individual characteristics in order to isolate the effect of informality status on earnings. Empirically, this has been achieved in a number of ways. In cases where only cross-sectional data is available, studies have controlled for selection either through Heckman-style correction procedures or instrumental variables (IV). Both approaches follow a similar two-step procedure. In the first stage, the endogenous variable – informality status – is modelled as a function of variables which are correlated with sectoral choice but uncorrelated with earnings. The predicted residuals are then used as an explanatory variable in the wage equation. Heckman correction procedures have been used by a number of studies, such as Heckman and Hotz (1986) for Panama, and Marcouiller, de Castilla and Woodruff (1997) for three Latin American countries. More recently, Mattos and Ogura (2009) used the proportion of pensioners per state to instrument for formal sector employment in Brazil. A common finding in these types of studies is that the earnings gap decreases substantially once selection is accounted for. Thus, segmentation only explains part of the differential that has been observed in the literature.

Instead of standard wage regressions, a number of studies have relied on matching techniques to strengthen the case for causality. Methods such as propensity score matching (PSM) ensure that earnings comparisons are based only on formal and informal workers who share identical characteristics. Pratap and Quintin (2006) used this approach to measure the informal earnings penalty in Argentina. Contrary to their regression estimates, they did not find any evidence of a formal sector wage premium. A similar study undertaken by Badaoui, Strobl, and Walsh (2008) for South Africa initially showed a gross earnings penalty for informal workers. However, the gap disappeared once formal sector taxes were accounted for. Despite the merit of matching techniques, it should be noted that the results are based on a select group of individuals who are not necessarily representative of the informal sector. It is precisely for this reason that Fields (2005) cautions against restricting the analysis to a sub-sample of workers.

An alternative way of looking at segmentation is to imagine what informal workers would earn if they were formally employed. If these individuals can earn more given their characteristics, then they are not maximizing their incomes as predicted by the theory of comparative advantage (Gunther & Launov, 2012). To test this formally, a number of studies have utilized counterfactual distributions to compare workers' actual and hypothetical earnings. This has typically been achieved using decomposition techniques, such as the Blinder-Oaxaca methodology and its extensions. This approach breaks down the earnings gap into two distinct components – (i) an endowments effect, and (ii) a coefficients effect. The endowments effect captures the portion of the earnings gap arising from differences in productive characteristics, while the coefficients effect captures differences in the returns to these characteristics. In the context of informal employment, this component has typically been attributed to segmentation. Following this approach, Rand and Torm (2012) and Dasgupta, Bhula-or and Fakthong (2015) found that most of the earnings gap is accounted for by differences in characteristics, which they interpret as evidence for the competitive labour market model.

Instead of relying on hypothetical scenarios, one could observe the actual earnings of workers who have held both formal and informal jobs. This allows us to examine whether changes in informality status is generally associated with an increase or a decrease in earnings. This is the main strategy of fixed effects (FE) models, which rely on inter-sector movers. Because FE operates on changes within the individual, the effect of time-invariant unobserved characteristics is also eliminated. For these reasons, fixed effects models are fairly standard in studies with panel data (e.g. Cho & Cho, 2011; Tansel & Kan, 2012; Bargain & Kwenda, 2014). In general, the results are consistent with other studies which find that differences in characteristics account for a sizable share of the earnings gap. It should be noted, however, that FE models assume sectoral transitions to be exogenous or random. This would be violated if workers only move into sectors where they expect their earnings to increase (Nordman, Rakotomanana, & Roubaud, 2016; Bargain & Kwenda, 2014). In this case, wages and sectoral choice should be modelled simultaneously to avoid bias (e.g. Magnac, 1991; Carneiro & Henley, 2002).³

The studies reviewed up to this point operate under the assumption that informal workers are homogeneous, in which case comparisons of average earnings are sufficient. However, this conceals substantial differences in terms of preferences, skills, and outcomes. To account for heterogeneity, some studies have disaggregated informal workers into sub-groups based on their employment status (e.g. Perry, et al., 2007; Nordman et al., 2016). This approach has showed, quite consistently, that self-employed types fare better than informal wage workers. In particular, their earnings are roughly similar – and in some cases, even greater – to those of formal workers (Arias & Khamis, 2008). In contrast, informal

³ This does not appear to be necessary in the present study as sectoral moves are not systematically related to either wage gains or losses. See section 6 for more details.

salaried workers usually face a substantial earnings penalty (Perry, et al., 2007). As a result, self-employment is typically associated with the upper-tier of the informal sector, while wage or salary work is associated with the easy-entry tier.

Although the above approach provides a more nuanced picture of informal workers, it ignores other forms of heterogeneity which are not associated with employment status. In order to capture this, some studies have turned to latent class methods. One of the earliest studies of this kind was conducted by Cunningham and Maloney (2001) to test for dualism among Mexico's microenterprises. Using factor and cluster analysis, they found that only a small fraction of firms exhibited features of segmentation. Gunther and Launov (2012) turned to finite mixture methods to identify the number and size of latent segments within Cote d'Ivoire's informal sector. They found two distinct groups of informal workers, each accounting for roughly half the size of the sector. To test for segmentation, the authors drew upon the approach of previous studies by comparing actual and hypothetical earnings distributions. In doing so, they found that informal workers in the easy-entry segment were earning much less than they could if they were formally employed. Applying the same methodology for the Turkish labour market, Ben Salem and Bensidoun (2012) observed that the vast majority of informal workers were not maximizing their potential earnings, leading them to reject the competitive labour market model.

A more direct way of dealing with heterogeneity is by examining wage gaps across the entire distribution. This has previously been achieved using conditional quantile regressions (CQRs). This method compares the earnings of formal and informal workers who are in the same quantile of their conditional distribution. Using this approach for Brazil, Tannuri-Pianto and Pianto (2008) observed that the earnings gap is largest at low levels of income and smallest at the top. Similar patterns have been observed in other countries such as Colombia (Garcia, 2017), Mexico, and South Africa (Bargain & Kwenda, 2014). In relation to Fields' (1990) model, informal workers at the lower end of the distribution comprise the easy-entry tier while those at the top constitute the upper tier. Perry and co-authors (2007), however, pointed out the potential bias associated with CQR. Specifically, the assumption that informal workers will retain their position in the formal sector earnings distribution contradicts the principle of comparative advantage. We discuss this in detail in the next section and propose an alternative approach which, to the best of our knowledge, has not yet been applied in this context.

From a policy standpoint, the gap in knowledge lies in identifying the characteristics of workers facing segmentation. In other words, who are the workers in Field's easy-entry tier? At present, large-scale quantitative studies offer little evidence to address these policy questions. Bargain and Kwenda (2014), for example, ran a probit analysis of workers at the top quintile of the informal sector. Their analysis showed that these individuals were more likely to be located in economically active areas and to be employed in larger enterprises. Not surprisingly, they also had higher levels of human capital as shown by their educational attainment and managerial positions. In Section 5, we examine other socio-demographic predictors which can be used to identify highly disadvantaged informal workers.

4 Data and methodology

4.1 Defining informal employment

The International Labour Organisation (ILO) defines informal workers as all individuals who "are not subject to national labour legislation and income tax or entitled to social protection and employment benefits" (ILO, 2013b, p.4). This definition captures the essence of informal employment but is generally

difficult to operationalize. For example, national labour force surveys which are the most common source of data for informal employment, do not usually include detailed questions on taxes or benefits. Consequently, many developing country studies have used employment status as a proxy for informality. In particular, self-employed and casual wage workers are usually considered informal, while wage or salary employees are treated as formal (Loayza & Rigolini, 2011; OECD, 2009). Though imperfect, it is “typically a good proxy for informality” (La Porta & Shleifer, 2014, p. 111).

In this study, we define informal workers based on the Proxy 1 definition provided by Indonesia’s Central Bureau of Statistics (BPS) (Appendix Figure 1).⁴ Proxy 1 uses a combination of employment status and primary occupation. Specifically, it identifies the following individuals as informal workers: (i) own account workers and casual workers either in agriculture or non-agriculture whose main occupation is sales worker, services worker, agricultural worker, animal husbandry, forestry worker, fisherman, hunter, production worker, transport equipment operator, labourer, or others; (ii) employer assisted by temporary or unpaid workers whose main occupation is agricultural worker, animal husbandry, forestry worker, fisherman, hunter, or others; and (iii) all family workers⁵. Compared to Proxy 2, this definition is more restrictive since it excludes workers in highly-skilled occupations (i.e. managers and professionals). This lessens the possible bias associated with defining informality solely on the basis of employment status.

4.2 Data

The Indonesian Family Life Survey (IFLS) is a multi-topic longitudinal survey which began with over 7,000 households and more than 30,000 individuals representing about 83 percent of the country’s population (Peterson, 1995). It provides two measures of labour market earnings, which serves as the dependent variable in this study. The first measure is based on earnings in the past year, while the second is based on earnings in the previous month. As we are primarily interested in the permanent component of income, we utilize annual earnings and convert it into real terms using the Consumer Price Index (CPI).⁶ We then construct the main dependent variable, hourly earnings, using real annual earnings, hours worked per week, and weeks worked per year. For the regressions, we transform hourly earnings into its log form so that the earnings gap is measured as a percentage of formal earnings rather than as an absolute difference. This normalization not only allows the gap to be comparable across quantiles, but also accounts for the relativity of earnings. That is, a small absolute gap matters more for workers at the bottom versus those at the top of the income distribution.

⁴ Other studies have defined informal employment using other definitions, such as social security contributions or benefits (e.g. Pratap & Quintin, 2006; Badaoui et al., 2008) and contract status (e.g. Tannuri-Pianto & Pianto, 2008). However, these measures have their own weaknesses. For example, labour contracts primarily apply to wage workers or employees, who represent a relatively small fraction of the working population. Moreover, written contracts may not be common even among workers who are considered formal. In Indonesia, for example, only about 20 percent of formal workers claim to have a labour contract (World Bank, 2010). In addition, more than 80 percent of all workers report not being covered by any pension scheme. This includes even public sector employees, who are considered to be core members of the formal sector.

⁵ While unpaid family workers are considered informal, they are excluded from the empirical analysis as they do not receive monetary compensation for their services.

⁶ Compared to annual earnings, monthly earnings are expected to be more volatile as it captures both the transitory and permanent components of income. This issue is likely to be worse for informal workers, whose earnings are less stable than formal workers.

Our main variable of interest relates to informality status, which we construct following the BPS Proxy 1 definition. The choice of other independent variables is based on the Mincer equation, which models earnings as an increasing function of education and experience. Because labour market experience is not explicitly measured in the survey, we rely on age as the main proxy. We also utilize tenure in the current job as an additional control for experience. In addition, we control for a number of other factors which are correlated with earnings such as gender, marital status, number of children in household, geographical location (i.e. urban vs rural), firm size, and sector of employment. Finally, we include both province and year effects. This model is estimated using the three most recent rounds of the IFLS, which were conducted in 2014, 2007, and 2000.

4.3 Methodology

We employ a number of empirical strategies to distinguish between competing models of informal employment. We begin our analysis by estimating the above earnings function using a pooled OLS regression clustered at the individual level. This is key to establishing: (i) whether there is a significant gap in the earnings of formal and informal workers; (ii) the size of the earnings gap; and (iii) whether the gap favors formal workers over informal workers. While we include controls for human capital (i.e. educational attainment, labour market experience) in the model, there are admittedly unobserved individual characteristics (e.g. unmeasured ability) which OLS cannot account for. To address this, we will supplement the analysis with a fixed effects estimation of the same model. In mathematical terms, this will take the following form:

$$Y_{it} - \bar{Y}_i = \alpha + \beta(X_{it} - \bar{X}_i) + \gamma(Z_i - \bar{Z}_i) + (\varepsilon_{it} - \bar{\varepsilon}_i) + (u_i - \bar{u}_i)$$

where X_{it} and Z_i are vectors of time-variant and time-invariant variables, respectively, ε_{it} is the stochastic error term, and u_i is an individual-specific error term capturing unobserved characteristics. Because FE models rely on mean-differencing, both observed (Z_i) and unobserved (u_i) time-invariant variables drop out of the equation. Consequently, the bias associated with time-invariant unobserved heterogeneity is eliminated. The combination of these two methods will help determine the nature of the earnings gap. If it disappears in the fixed effects estimation, then the gap is caused by unobserved individual characteristics as suggested by the theory of comparative advantage. Once time-invariant factors are controlled for, the remaining gap may capture either compensating differentials or segmentation.

As mentioned earlier, however, neither pooled OLS nor OLS-FE models can account for heterogeneity in earnings since they can only estimate gaps at the mean. More advanced models are thus needed to tease out differences in the size of the formal-informal earnings differential. Previous studies have overcome this by running conditional quantile regressions which were first introduced by Koenker and Basset (1978). This method creates separate distributions based on the covariates in a given model. It then compares the outcomes of two individuals who are in the *same* quantile of their conditional distribution. Consider, for example, a simplified model of earnings in which the only covariate is a dummy variable for formal employment. This will create two separate distributions – one for formal workers, and another for informal workers. CQR estimates the returns to formal employment by comparing the earnings of formal and informal workers who are at the *same* quantile of their conditional distribution. If the median earner in the formal sector earns twice as much as the median earner in the informal sector, then we can conclude that moving to the formal sector is associated with a doubling of income for workers at the middle of the income distribution.

Clearly, there are some problems with this type of analysis. For one, it is not always reasonable to assume that workers will remain in the same quantile given a slight change in their characteristics (Koenker, 2005). Importantly, this assumption is likely to be violated in comparisons of formal and informal work. As the theory of comparative advantage implies, workers choose the occupation or sector that will maximize their skills or endowments. Thus, an informal worker at the 80th percentile of his or her conditional distribution may not necessarily remain in that income quantile when he or she transitions to a formal job (Perry, et al., 2007). Moreover, results from CQRs are highly sensitive to model specification because the conditional distributions are constructed based on the covariates in the model (Lass & Wooden, 2017). This has two important implications. First, it would be difficult to compare findings across studies that utilize different regressors. And second, any policy implication that may be derived from CQRs would apply only to a subset of the population (Firpo et al., 2009).

To overcome these limitations, we will follow the unconditional quantile regression (UQR) approach proposed by Firpo et al. (2009). Unlike its predecessor, the UQR only considers one distribution – the overall earnings distribution – which is pre-determined and does not depend on the model. This method estimates the change in earnings at each quantile following a marginal change in one of the covariates. In the context of this study, UQR will allow us to estimate the effect of a 1 percentage point increase in informal employment on the earnings of those at the lowest quantile (or any quantile for that matter). In this regard, differences in the magnitude of the change in earnings would be consistent with heterogeneity among informal workers.

Econometrically, the method builds on the influence function (IF). As the name suggests, the IF represents the influence of an individual observation on a distributional statistic (Firpo et al., 2007). For example, the influence function of the mean is simply the demeaned value of an observation ($y_i - \mu$). In terms of quantiles, the IF is given by:

$$IF(y, q_\tau) = \frac{(\tau - I\{Y \leq q_\tau\})}{f_y(q_\tau)}$$

where I is an indicator function, $f_y(\cdot)$ is the density of the marginal distribution Y , and $q_\tau = Q_\tau[Y]$ is the population τ -quantile of the unconditional distribution Y . Adding back the distributional statistic to the influence function generates the Recentered Influence Function (RIF):

$$RIF(Y, q_\tau; F_y) = q_\tau + \frac{\tau - I\{Y \leq q_\tau\}}{f_y(q_\tau)}$$

Thus, the RIF for quantiles can be expressed simply as $q_\tau + IF(y, q_\tau)$. In the actual regression, the RIF serves as the outcome variable, replacing the traditional dependent variable, y_i . By modelling the RIF as a function of the vector of explanatory variables X_i , we are able to quantify the change in any earnings quantile associated with a change in informality status.

5 Results and discussion

The 2000, 2007, and 2014 waves of the IFLS contain a total of 73,321 observations which correspond to 40,173 unique individuals. For the analysis, we focus only on individuals aged 15 to 60 throughout the sample period. Furthermore, we limit the analysis to those who reported positive labour

earnings. These restrictions reduce the sample to 39,979 observations or 24,692 individuals. Of these, 17,123 individuals or 69 percent of the sample have observations in at least two waves. Table 1 provides summary statistics for formal and informal workers based on the sample.

Table 2 provides the raw earnings gap at the mean and at different points in the distribution. On average, informal workers earn 11 percent per hour compared to formal workers. While the mean earnings gap does not appear to be as substantial as the literature suggests, it should be noted that it represents the average across three waves (i.e. 2000, 2007, 2014).⁷ More importantly, Table 2 suggests that the earnings gap varies across the distribution. In particular, the gap appears largest at the bottom and declines steadily as incomes increase. The difference in the earnings gap between the 10th and 90th percentile is also quite significant. Informal workers at the top of the distribution face an earnings penalty that is 13 percentage points less than those at the bottom.

In order to control for the effect of observable characteristics, we first estimate a pooled OLS regression for the full sample and for men and women separately (Table 3). The results are generally in line with the literature. The coefficients on age suggest that earnings increase at a decreasing rate as people get older. The dummy variable for women is negative and significant, confirming the usual observation that women earn less than their male counterparts. Workers located in rural areas earn less, on average, than those in urban areas. Furthermore, the coefficients of education and job tenure are both highly significant and follow the expected signs. Specifically, earnings increase as individuals attain higher levels of schooling and accumulate more tenure on the job. Most importantly, the results from the pooled OLS regression confirm the existence of an earnings penalty for informal workers. On average, these workers earn around 19 percent less than their formal counterparts. However, the gap appears to be significantly larger for men than women.

As mentioned earlier, the earnings gap may arise for a number of reasons. One possible explanation is related to sorting or selection of workers based on time-invariant unobservable characteristics. In order to eliminate its effect on the earnings gap, we supplement the pooled OLS regression with a fixed effects regression on the same empirical model. The results based on the full sample shows that the estimated earnings gap is reduced by nearly half once time-invariant unobserved factors are eliminated. Interestingly, this result appears to be driven primarily by male workers as the coefficient for women is small and statistically insignificant.

Because OLS and fixed effects regressions can only estimate the earnings gap at the mean, they are unable to capture the heterogeneity of earnings among informal workers. To address this limitation, we run the unconditional quantile regression proposed by Firpo et al. (2007, 2009). Figure 1 and Table 4 show UQR estimates of the informal earnings gap at different quantiles of the distribution. The trend is broadly consistent with the descriptive analysis presented in Table 2. That is, the earnings gap is largest at the bottom of the distribution and decreases non-monotonically as it approaches the top quantile. The size of the gap based on the full sample ranges from as high as 30 percent for workers at the 10th percentile of earnings to as low as 8 percent for those at the 90th percentile. Notably, the magnitude of the earnings penalty is larger for men than women across the distribution, although the same pattern can be observed

⁷ Incidentally, these three years capture very different macroeconomic conditions which have a direct impact on the size and nature of formal and informal work. In particular, the year 2000 was characterized by a slowing down of the Indonesian economy as it reeled from the 1997 Asian Financial Crisis. During that time, informal employment grew as many workers were displaced from their formal jobs. For others, it also meant a decline in real earnings. Consequently, the earnings gap narrowed during that period. However, the gap widened once more in 2007 and 2014 as Indonesia's economy picked up. For these years, informal workers earn about 25 percent less than formal workers on average.

for both.

The main drawback of the UQR is its inability to eliminate time-invariant unobserved factors that may influence the earnings gap. Consequently, UQR estimates may overstate the penalty faced by informal workers. One way to resolve this is to combine UQR with fixed effects (UQR-FE). To implement this, we follow the extension proposed by Borgen (2016).⁸ Two things are worth noting from the results shown in Table 5 and Figure 2. Similar to the earlier findings, UQR-FE estimates show that the earnings gap declines as income levels increase. Thus, workers at the bottom of the distribution face the largest earnings penalty while those at the top face little to no penalty at all. The main difference, however, lies in the magnitude of the earnings gap. Across all quantiles, UQR-FE estimates are significantly smaller than UQR estimates particularly among men. Because the former eliminates time-invariant unobserved characteristics, the remaining gap may be interpreted as evidence for segmentation.

Heterogeneity in the size of the earnings gap is consistent with Fields' (1990, 2005) model of dualism among informal workers. This has clear implications for policy. For one, it implies that not all informal workers are better off in formal jobs. Therefore, policies which are designed to facilitate transitions from informal to formal employment need to be targeted towards those who are more likely to face segmentation. As the previous section showed, these individuals are concentrated at the bottom of the earnings distribution. A simple probit analysis of these workers can help identify their characteristics.⁹ As Table 6 shows, women are more likely to be at the bottom of the informal earnings distribution than men. Bottom income earners are also less likely to be married and more likely to be located in rural areas. Consistent with this latter finding, they are also more likely to be employed in agriculture. As expected, bottom earners also have lower levels of human capital as shown by their low educational attainment and shorter job tenure.

6 Robustness checks

6.1 Intersector movers as the basis of identification

While FE models are fairly standard across the literature, several conditions need to be satisfied for it to work. For one, there needs to be sufficient movement across the formal and informal sectors. Appendix Table 1 shows the number and share of intersector movers between 2000 and 2014. Overall, movers accounted for close to 30 percent of all workers.

Another issue relates to the use of intersectoral movers as the basis of identification. As some studies have pointed out, movers could be distinct from the stayers in the sample (e.g. Badaoui et al., 2008; Nordman et al., 2016). For example, individuals may shift from an informal to a formal job if they expect to earn more as a formal worker. If this were the case, then transitions from informal to formal employment should be systematically related to increases in earnings. That is, the vast majority of workers who make these transitions should experience positive earnings growth. However, a simple tabulation of earnings growth by transition type shows that this is not the case. Of the 1,769 observations corresponding to informal to formal transitions, only 1,114 or a little over 60 percent was associated with an increase in earnings while the rest was associated with a reduction in earnings. On the other hand, of

⁸ This is implemented using the *xtrifreg* command in STATA.

⁹ The dependent variable in this case takes on a value of 1 if the worker is in the bottom decile/quintile/tercile of the informal earnings distribution.

the 2,413 observations corresponding to formal to informal transitions, only 1,029 or 43 percent was associated with a decline in earnings while the remainder was associated with an earnings gain.

Another way to compare movers and stayers is to run a probit analysis of the two groups. The results show that movers have a number of distinguishing characteristics.¹⁰ Compared to stayers and the rest of the sample, movers are more likely to be male and generally older. They are also more likely to have low educational attainment and shorter job tenure. Movers are also more common in some sectors such as construction and wholesale and retail trade. All in all, however, the model estimates very little difference between movers and stayers in the sample. Specifically, the R-squared ranges from 0.07 to 0.08 depending on the control or reference group. These figures are generally in line with the R-squared obtained by other authors (e.g. Bargain & Kwenda, 2014; Nordman et al., 2016), which ranges from a minimum of 0.01 to 0.06.

6.2 The role of rural-urban migration

It is also important to consider the role of rural-urban migration in explaining the earnings differentials between formal and informal workers. Indeed, it may be argued that the earnings gap arises not from sectoral transitions but from changes in geographical location. For example, moving from an informal to a formal job may require relocating from a rural to an urban area. If jobs in rural areas pay less than those in urban areas, then workers who migrate will receive higher earnings. Alternatively, transitions from formal to informal employment may involve migration out of urban centers and into rural areas. In both scenarios, the earnings gap associated with intersectoral transitions could simply be capturing the effect of migration.

To determine whether the two are indeed related, we cross-tabulate moves across sectors and geographical location (Appendix Table 3). The figures do not show a systematic relationship between the two. Of the 2,413 observations corresponding to formal-informal transitions, about 42% remained in rural areas (Rural-Rural) and about 43% remained in urban areas (Urban-Urban). In terms of informal-formal transitions, around 39 percent of 1,769 observations remained in rural areas (Rural-Rural) and 46 percent remained in urban areas (Urban-Urban). Additionally, we examine the pairwise correlation between the two variables. Not surprisingly, this turns out to be very low: -0.09 to be exact.

6.3 Distinguishing compensating differentials from segmentation

As mentioned earlier, the use of earnings to test for segmentation has been criticized on the basis of compensating differentials. Put simply, the existence of an earnings gap is not a sufficient condition for segmentation because lower earnings may be offset by non-pecuniary aspects of the job. To address this, Perry and co-authors (2007) suggested examining individuals' perceptions of their own welfare. Workers facing segmentation in the labour market should not only receive lower earnings, but also report lower levels of welfare compared to formal workers. In contrast, individuals who engage in informal work for its non-monetary benefits should report similar levels of welfare compared to their formal counterparts.

To address this, we follow the approach of Arias and Luchetti (2007) who examined differences in self-rated poverty measures between formal and informal workers in Argentina and the Dominican Republic. Specifically, we use data from the subjective wellbeing module of the IFLS where respondents are asked to identify their position in the economic ladder. Individuals rate themselves on a scale of 1 to

¹⁰Results are available upon request.

6, with the lowest value representing “poorest” and the highest representing “richest”. In order to identify self-rated poverty, we create a dummy variable which takes on a value of 1 if the respondent places themselves at step 1 or 2 of the economic ladder. We then run a simple probit analysis of self-rated poverty, where the independent variables include informality status.¹¹ In order to control for income, we limit the sample to workers at the bottom decile of the earnings distribution. The results show that informal workers are significantly more likely to rate themselves as poor compared to their formal counterparts.¹² This finding appears consistent with segmentation at the bottom.

7 Conclusion and policy implications

Unlike many other empirical studies, this paper directly addressed the concept of heterogeneity among informal workers. Using unconditional quantile regressions, we provided evidence for Fields’ (1990) model of an informal sector composed of an upper tier and an “easy-entry”, lower tier. Workers in the upper tier earn comparable levels of income to formal workers, while those in the lower tier have significantly lower earnings than their formal counterparts. We also conduct additional tests to identify the most likely explanation for the earnings penalty at the bottom of the distribution. By combining fixed effects with UQR, we eliminate the possibility that the gap arises from time-invariant unobserved characteristics consistent with comparative advantage. We also perform a robustness analysis using self-rated poverty to account for the role of compensating differentials. The results confirm earlier theoretical arguments that the upper tier of the informal sector may indeed be desirable with regard to earnings.

The evidence presented thus far has some implications for policy. First, a “one-size-fits-all” solution would be inappropriate given the diversity of outcomes in the informal sector. Instead, policies need to be targeted towards groups that are more likely to suffer from segmentation. The probit analysis performed earlier provides an example of how these groups can be identified. Second, the possible existence of a competitive segment within the informal sector contradicts the notion that all informal jobs are “bad jobs”. If informal employment is optimal for some individuals, then policies aimed at reducing it may need to be reconsidered.

Like the studies reviewed earlier however, this article is limited by its focus on earnings. Admittedly, this approach ignores all other aspects of informal jobs which may contribute to an individual’s overall utility. Consequently, this study is unable to provide conclusive evidence for workers at the bottom of the earnings distribution. Addressing the issue of compensating differentials would require a more direct and comprehensive measure than what the earlier analysis has provided.

¹¹ The other independent variables are age, sex, marital status, geographical location, educational attainment, sector of employment, and job tenure. We also control for province and year effects, and cluster by individuals.

¹² The estimated coefficient is significant at the 95% level.

Table 1. Summary statistics (2000-2014)

Variable	Total		Q1		Q2		Q3		Q4	
	Formal	Informal	Formal	Informal	Formal	Informal	Formal	Informal	Formal	Informal
<i>Age (average, in years)</i>	34	38	32	38	32	38	33	38	37	38
Gender										
Male	63.1	71.4	48.3	60.3	65.9	71.3	70.0	76.2	68.2	77.7
Female	36.9	28.6	51.7	39.7	34.1	28.7	30.0	23.8	31.8	22.3
Marital status										
Single	22.3	9.4	29.1	10.2	26.1	9.1	21.0	10.7	12.9	7.8
Married	73.3	83.3	64.1	80.6	69.8	83.1	76.0	82.6	83.6	86.6
Separated/Divorced	2.6	3.4	4.0	4.0	2.6	3.9	1.9	3.1	2.0	2.6
Widowed	1.8	3.9	2.8	5.2	1.5	3.9	1.1	3.6	1.5	3.0
Average # of children in HH	1.62	1.75	1.73	1.76	1.60	1.73	1.56	1.77	1.60	1.73
Has child below 5 in HH	40.3	40.7	38.6	38.9	40.1	40.8	42.1	39.8	40.4	43.2
Geographic location										
Urban	68.1	40.8	59.5	30.6	65.5	39.8	71.8	46.8	75.6	46.2
Rural	31.9	59.2	40.5	69.4	34.5	60.2	28.2	53.2	24.4	53.8
Educational attainment										
Elementary	42.3	73.2	66.2	81.8	51.7	76.8	33.6	72.6	18.9	62.2
High school	35.7	21.7	25.1	15.4	36.5	20.2	45.9	22.3	34.9	28.6
College	19.8	4.0	7.4	1.7	10.6	2.5	19.0	3.9	41.5	7.8
Post graduate	1.0	0.0	0.0	0.0	0.2	0.0	0.3	0.1	3.5	0.1
Others (e.g. adult education)	1.2	1.1	1.2	1.1	1.1	0.5	1.3	1.1	1.2	1.4
Job tenure (<i>average, in years</i>)	6.6	9.0	5.0	9.4	5.6	9.1	6.4	8.8	9.5	8.7

Variable	Total		Q1		Q2		Q3		Q4	
	Formal	Informal	Formal	Informal	Formal	Informal	Formal	Informal	Formal	Informal
Sector of employment										
Agriculture	8.8	44.3	12.8	54.1	10.0	45.3	7.2	36.7	5.2	41.1
Mining and quarrying	1.1	0.8	0.6	0.6	0.8	0.6	1.3	1.0	1.7	0.9
Manufacturing	21.0	5.8	20.0	7.0	23.6	5.9	24.3	5.2	16.0	4.9
Electricity, gas, and water	0.7	0.1	0.3	0.0	0.5	0.1	1.0	0.2	1.0	0.1
Construction	4.9	6.7	4.3	2.4	7.0	5.0	5.4	11.2	3.1	8.4
Wholesale and retail, hotels and restaurants	23.6	24.8	31.0	24.0	24.8	24.2	21.4	25.2	17.1	25.8
Transportation, storage and communications	3.5	5.3	3.4	3.4	4.2	6.9	3.8	6.9	2.8	4.1
Finance, insurance, real estate and business	3.8	0.9	1.2	0.5	2.7	0.4	4.5	1.2	6.9	1.7
Social services	32.0	10.7	26.2	7.6	26.1	11.1	30.4	11.9	45.4	12.4
Activities not elsewhere classified	0.5	0.6	0.3	0.5	0.4	0.6	0.7	0.7	0.8	0.8

Note: Unless otherwise specified, figures represent shares to total.

Table 2. Raw earnings gap at various points of the distribution (2000-2014)

Quantile	Real hourly earnings (in IDR)		Raw earnings gap (as a share of formal earnings)
	Formal	Informal	
Average	12,684.96	11,257.72	11.3%
10 th percentile	1,262.63	721.50	42.9%
20 th percentile	2,180.53	1,286.17	41.0%
30 th percentile	3,084.67	1,874.89	39.2%
40 th percentile	4,058.44	2,525.25	37.8%
50 th percentile	5,303.46	3,349.41	36.8%
60 th percentile	6,919.47	4,416.81	36.2%
70 th percentile	9,264.13	5,864.01	36.7%
80 th percentile	12,806.57	8,106.14	36.7%
90 th percentile	20,706.71	14,395.12	30.5%

Note: The raw earnings gap is calculated as 1 minus the ratio of informal to formal earnings.

Table 3. Pooled and fixed effects estimates of the average earnings penalty (2000-2014)

Variables	Pooled OLS			Fixed effects		
	Full sample	Men	Women	Full sample	Men	Women
Informal	-.1864*** (.0162)	-.2253*** (.0187)	-.1109*** (.0304)	-.0954*** (.0249)	-.1117 *** (.0290)	-.0458 (.0498)
Age	.0434*** (.0046)	.0358*** (.0056)	.0439*** (.0079)	.0946*** (.0145)	.1066*** (.0165)	.0536* (.0311)
Age squared	-.0005*** (.0001)	-.0004*** (.0001)	-.0005*** (.0001)	-.0007*** (.0001)	-.0008*** (.0001)	-.0006*** (.0002)
Marital status (ref: Single)						
Married	.1647*** (.0207)	.2117*** (.0250)	.1637*** (.0352)	.0218 (.0432)	.0375 (.0507)	-.0219 (.0831)
Separated/Divorced	-.0669 (.0458)	.0946 (.0747)	.0302 (.0596)	.0904 (.0821)	.1047 (.1084)	.0518 (.1314)
Widowed	-.1009* (.0536)	.1232 (.0956)	.0675 (.0680)	-.0137 (.0948)	.1432 (.1484)	-.1487 (.1381)
Rural	-.0943*** (.0150)	-.0653*** (.0178)	-.1444*** (.0263)	-.0589* (.0348)	-.0382 (.0400)	-.1306* (.0705)
# of children in HH	-.0161*** (.0053)	-.0103 (.0064)	-.0130 (.0091)	-.0261*** (.0092)	-.0288*** (.0108)	-.0218 (.0174)
HH has child below 5	.0318** (.0142)	-.0071 (.0170)	.0101 (.0253)	.0099 (.0207)	-.0007 (.0241)	.0498 (.0409)
Educational attainment (ref: Elementary)						
High school	.4561*** (.0152)	.3724*** (.0179)	.5370*** (.0276)	.1317** (.0630)	.1535** (.0701)	.0226 (.1428)
College	.9915*** (.0213)	.9346*** (.0269)	1.1006*** (.0348)	.2893*** (.0858)	.3109*** (.0994)	.2045 (.1752)
Post graduate	1.697*** (.0651)	1.5865*** (.0847)	1.8692*** (.1041)	.5159*** (.1542)	.5829*** (.1874)	.3243 (.2806)
Others (e.g. adult education)	.2649*** (.0608)	.2460*** (.0702)	.2129*** (.1146)	.0346 (.0996)	.0509 (.1156)	-.0039 (.1984)

Job tenure	.0151*** (.0010)	.0113*** (.0012)	.0213*** (.0018)	.0058*** (.0014)	.0045*** (.0017)	.0089*** (.0029)
Sector controls	Yes	Yes	Yes	Yes	Yes	Yes
Province effects	Yes	Yes	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes	Yes	Yes
Constant	7.4595*** (.3349)	7.6342*** (.3086)	6.6318*** (.1374)	5.8857*** (.3702)	5.6775*** (.4160)	6.8960*** (.8117)
Observations	32,365	21,257	11,108	26,016	17,215	8,801
R-squared	0.2298	0.2209	0.2640	0.0748	0.0698	0.1382

*** p<0.01, ** p<0.05, * p<0.1

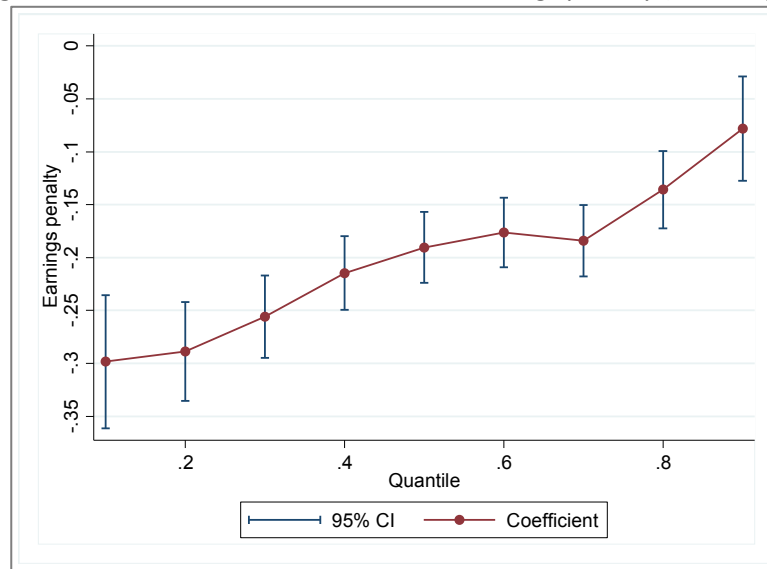
Table 4. UQR estimates of the informal earnings penalty (2000-2014)

Quantile	Full sample	Men only	Women only
0.10	-0.298*** (0.0321)	-0.390*** (0.0382)	-0.173*** (0.0589)
0.20	-0.289*** (0.0238)	-0.320*** (0.0269)	-0.163*** (0.0457)
0.30	-0.256*** (0.0198)	-0.275*** (0.0219)	-0.172*** (0.0383)
0.40	-0.215*** (0.0177)	-0.245*** (0.0204)	-0.140*** (0.0355)
0.50	-0.190*** (0.0171)	-0.224*** (0.0196)	-0.0809** (0.0335)
0.60	-0.176*** (0.0168)	-0.221*** (0.0194)	-0.0721** (0.0336)
0.70	-0.184*** (0.0172)	-0.212*** (0.0206)	-0.0787** (0.0332)
0.80	-0.136*** (0.0187)	-0.161*** (0.0228)	-0.0539 (0.0340)
0.90	-0.078*** (0.0252)	-0.0884*** (0.0325)	-0.0425 (0.0397)
Demographic controls	Yes	Yes	Yes
Education and employment controls	Yes	Yes	Yes
Province effects	Yes	Yes	Yes
Year effects	Yes	Yes	Yes
Observations	33,493	21,799	11,694

^a Robust standard errors in parentheses

^b Demographic controls include age, age squared, marital status, urban/rural status, number of children in HH, and a dummy for child in HH below the age of 5. Education and employment controls include the highest level of educational attainment, job tenure, and sector of employment. The full set of results are available on request.

^c *** p<0.01, ** p<0.05, * p<0.10

Figure 1. UQR estimates of the informal earnings penalty (full sample)**Table 5.** UQR-FE estimates of the informal earnings penalty (2000-2014)

Quantile	Full sample	Men only	Women only
0.10	-0.228** (0.0924)	-0.283*** (0.102)	-0.109 (0.219)
0.20	-0.163** (0.0698)	-0.183** (0.0737)	-0.115 (0.168)
0.30	-0.157*** (0.0580)	-0.185*** (0.0598)	-0.0715 (0.138)
0.40	-0.120** (0.0518)	-0.134** (0.0543)	-0.0679 (0.124)
0.50	-0.0841* (0.0507)	-0.0823 (0.0525)	0.0331 (0.120)
0.60	-0.0533 (0.0502)	-0.0661 (0.0523)	-0.0220 (0.121)
0.70	-0.0742 (0.0527)	-0.0833 (0.0560)	-0.0581 (0.124)
0.80	-0.000750 (0.0572)	-0.0224 (0.0619)	0.0361 (0.128)
0.90	-0.0216 (0.0780)	-0.0598 (0.0899)	0.0855 (0.152)
Demographic controls	Yes	Yes	Yes
Education and employment controls	Yes	Yes	Yes
Province effects	Yes	Yes	Yes
Year effects	Yes	Yes	Yes

Observations	26,972	17,664	9,308
# of pidlinkds	16,161	9,909	6,252

^a Standard errors in parentheses

^b Demographic controls include age, age squared, marital status, urban/rural status, number of children in HH, and a dummy for child in HH below the age of 5. Education and employment controls include the highest level of educational attainment, job tenure, and sector of employment. The full set of results are available on request.

^c *** p<0.01, ** p<0.05, * p<0.1

Figure 2. UQR and UQR-FE estimates of the informal earnings penalty (full sample)

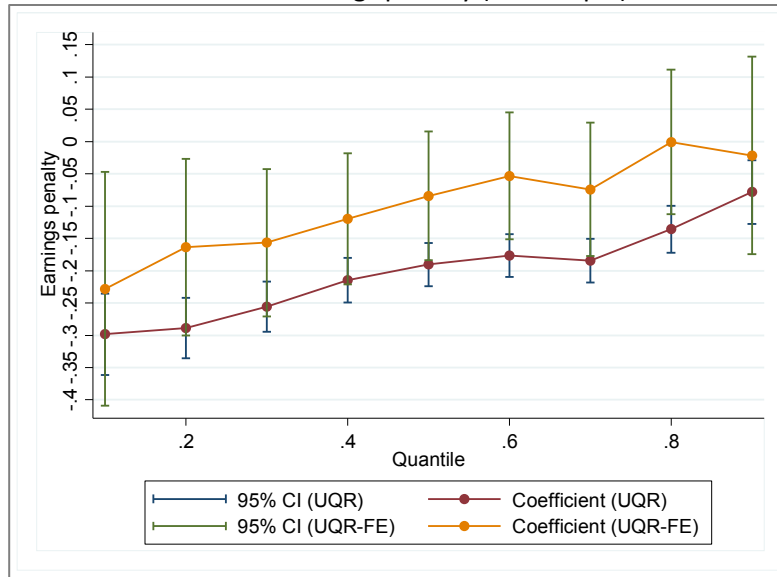


Table 6. Probit analysis of informal workers at the bottom of the earnings distribution

Variables	(1) bottom 10%	(2) bottom 20%	(3) bottom 30%
Age	0.00247 (0.00228)	0.00226 (0.00191)	0.00144 (0.00175)
Female	0.378*** (0.0431)	0.433*** (0.0368)	0.437*** (0.0347)
Marital status (ref: single/not married)			
Married	-0.238*** (0.0643)	-0.269*** (0.0534)	-0.231*** (0.0496)
Separated/Divorced	-0.105 (0.115)	-0.148 (0.0970)	-0.137 (0.0902)
Widower	-0.238*** (0.0643)	-0.269*** (0.0534)	-0.231*** (0.0496)
Rural	0.154*** (0.0436)	0.234*** (0.0368)	0.201*** (0.0338)

Educational attainment (ref: elementary)			
High school	-0.302*** (0.0517)	-0.269*** (0.0408)	-0.233*** (0.0364)
College	-0.331*** (0.111)	-0.537*** (0.0948)	-0.619*** (0.0836)
Post graduate	-	-	-
Others	0.143 (0.160)	0.0960 (0.148)	-0.0693 (0.137)
Tenure	-0.00771*** (0.00252)	-0.00608*** (0.00196)	-0.00539*** (0.00178)
Sector of employment (ref: agriculture)			
Mining and quarrying	-1.021** (0.403)	-0.275 (0.201)	-0.408** (0.185)
Manufacturing	-0.377*** (0.0802)	-0.224*** (0.0644)	-0.214*** (0.0594)
Electricity, gas, and water	-	-	-
Construction	-0.495*** (0.0926)	-0.647*** (0.0796)	-0.645*** (0.0689)
Wholesale and retail, restaurants and hotels	-0.358*** (0.0496)	-0.324*** (0.0427)	-0.333*** (0.0401)
Transportation, storage, and communications	-0.693*** (0.116)	-0.531*** (0.0811)	-0.423*** (0.0689)
Finance, insurance, real estate and business	-0.234 (0.216)	-0.111 (0.177)	-0.333** (0.170)
Social services	-0.483*** (0.0735)	-0.396*** (0.0581)	-0.313*** (0.0523)
Activities n.e.c.	-0.208 (0.223)	-0.398* (0.204)	-0.110 (0.173)
Establishment size	-0.000761 (0.000638)	-0.00243* (0.00145)	-0.00273** (0.00121)
Province effects	Yes	Yes	Yes
Year effects	Yes	Yes	Yes
Constant	-0.916*** (0.122)	-0.553*** (0.103)	-0.447 (0.631)
Observations	10,132	10,160	10,175
Pseudo R-squared	0.0752	0.0827	0.0777

Note: Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Appendix Figure 1. BPS definitions of informal employment

Employment Status	Main occupation									
	Professional	Managerial	Clerical	Sales	Services	Agricultural workers	Production	Operational	Laborer	Others
Own account worker										
Employer assisted by temporary worker										
Employer assisted by permanent worker										
Employee										
Casual employee in agriculture										
Casual employee in non-agriculture										
Unpaid family worker										

	Proxy 2
	Proxy 1 & 2

Source: Adapted from Badan Pusat Statistik Indonesia

Appendix Table 1. Share of intersector movers to all workers (2000-2014)

	Informal	Formal
Informal	9,202 68.8%	4,177 31.2%
Formal	3,651 27.6%	9,575 72.4%

Appendix Table 2. Moves across sectors and geographical location

Sectoral transition	Rural-urban migration			
	Rural-Rural	Rural-Urban	Urban-Rural	Urban-Urban
Formal - Formal	1,699	728	224	4,858
Formal - Informal	1,005	252	122	1,012
Informal - Formal	693	220	39	807
Informal - Informal	2,206	273	72	947

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